

Information Technology Education-AAMUSTED

COURSE MANUAL

COURSE INFORMATION

Course Title: DATABASE CONCEPTS AND DESIGN

Course Code: ICT 231

Credit Hours: 3 Credits

INSTRUCTOR'S INFORMATION

Name: Boansi, Oliver Kufuor

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HOURS: To Be Communicated later

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COURSE DESCRIPTION

Databases are very important in this era where information is key in major organizational decisions. Badly designed databases stem out of the fact that key concepts and principles required to engineered well-structured databases are lacking.

The course is designed to expose students to the essential database concepts, technology and techniques necessary to begin a career as a database developer. It is intended to equip the student with sufficient knowledge and skills of concepts that underpins the development of database systems and modern DBMS practices.

TEXTBOOK & COURSE MATERIALS:

Main Textbook:

Coronel, C. et al (2010). Database systems design, implementation and management (9th ed.), U.S.A : Cengage Learning.

Supplementary Books:

Conger, S. (2012). Hands-on database: An Introduction to Database Design and Development. Prentice Hall.

Kroenke, A. (2006). Database processing (10th Ed.), Boston: Prentice Hall.

Hoffer, J.A., Ramesh V., and Topi, H. (2013). Modern database management (11th ed.), Boston: Pearson.

Kroenke, D. & Auer, D. (2010). Database concepts. Boston: Prentice Hall.

Elmasri, R., Navathe, S., B. (2011). Fundamentals of database systems. London: Addison Wesley

Mannino, V., M., Database design, application development, & administration. (2004).

COURSE CONTENT

Topics discussed in this course includes overview of DBMS, purpose of Database systems, Enterprise and personal database systems, data models, relational model, relational algebra & Various operations, Introduction to SQL, E-R Diagram, Mapping, Constraints, Keys, Normalization, Database design issues, transaction concepts, Database Administrator, Introduction to Data mining and Data warehousing, etc.

CONNECTION WITH OTHER COURSE(s)

Databases are key in most if not all organizations because it is the base that informs management when it comes to making decisions that enhances their competitive edge over competitors. This course must be taken with all seriousness because it will aid the student in progressing to take and apply the knowledge and skills gained in courses like Database Development and Implementation, Visual programming, Web Technologies amongst other courses.

STUDENT ACTIVITIES/INSTRUCTIONAL STRATEGIES

- **Assignments and Quizzes:** There will be several assignments and quizzes due at specific dates. These assignments and quizzes will take the forms of written and practical.
- **Test(s):** There will be one major test in class at a scheduled date.
- **Examination(s):** There will be a written final exam.
- No incomplete grades will be given in this class.
- Poor attendance will result in a lower grade. If a student has a genuine reason for being late or absent, he/she is expected to notify the instructor ahead of time (before the date of absence or lateness). Failure to do this may prevent the student from making up a missed quiz or exams.

STUDENT LEARNING OUTCOME

By the end of this course, a student should be able to:

- Describe the importance of databases and its design
- Model data using ERD as a data modelling tool
- Apply normalization rules in the design of a database
- Explain the concepts underlining the design of databases

- Perform basic SQL queries
- Identify the components of a database management system
- Outline the steps of database design

TOPIC OUTLINE/SCHEDULE

WEEK	TOPIC	LEARNING ACTIVITIES	PRE-LESSON PREPARATION	DUE DATE
1	Database Systems • The Need for Databases • Data Versus Information • Introduction to Database (Forms, Types of databases) • Evolution of file System Data Processing • Problems with File System Data Processing • Importance of Database Design • The Database System Environment • DBMS Functions • Role and Advantages of DBMS	• Introduction to course manual usage. • Instructor leads out in discussing topics by asking relevant questions • Assignment 1 is given	Read ch.1 of the main textbook	
2	The Database Life Cycle ACID Properties of a transaction	•		Assignment 1
3	Data Models • Data Modeling and Data Models • Importance of Data Models • Data Model Building Blocks • Relationships within Relational Database (1:M, 1:1, M:N) • Business Rules (Discovering Business Rules) • Translating Business Rules into Data Model Components		Read Ch. 2 of textbook	Assignment 1 due
	• Naming Conventions • Evolution of Data models • Degrees of Data Abstraction			

4	- Entity Relationship Modeling Entities, - Attributes, Relationships - Connectivity and Cardinality - Existence Dependence, Relationship Strength, - Weak entities Relationship Participation, Degree Recursive Relationships Associative (Composite) Entities		Read Ch. 4 of Main Textbook	Quiz 1
5	Developing an ER Diagram	- A description of the process is demonstrated to student - Students take turn to develop models for problems given in groups	- Read Ch.4 of the main textbook - Install the data modelling tool on laptop	
6	Relational Database Model - Keys - Integrity rules - Indexes Logical Design (Converting ERDs to Relations)		Read Ch. 3 of main textbook	Assignment 2 due
7	Normalization of Database Relations - The Need for Normalization - Functional Dependency - Normalization Process (1NF, 2NF, 3NF)		Read Chap.4 of Hoffer's book.	
8	Relational Algebra (Set Operators) Union, Intersection, Difference, Selection, Projection, Joining and Division	Instructor introduces the concepts with examples. Students take turn to practice the concepts, coming out with their examples.	Re-Read Ch.3 of main text book	

9	Modern Database Systems • NoSQL Databases, Big data technologies, cloud databases, Data warehousing		Read Unit 13 of the course material	
10	Introduction to SQL • Data Definition Commands • Data Manipulation Commands SELECT Queries		Read Ch.7 of main textbook. In addition read Elmasri's Ch.4	
11	SQL Introduction continued Introduction to Data Mining Introduction to Data Warehouse		Quiz 2	Submission of term's project
12	Group Presentations on Assigned Topics			
13	Group Presentations on Assigned Topics			
14	Overview of the entire Course			

GRADING POLICY

GRADE EVALUATION

Assignments/Quizzes	5%
Mid-Sem. Exam	15%
Final Exam	60%
Term Project/Presentation	15%
Class Attendance / Participation	5%

LATE WORK POLICY

Late assignments are discouraged and may attract point deductions or may not be accepted for grading. There will be no make-up assignments or quizzes or late work accepted without a serious and compelling reason and lecturer approval.

GRADING SYSTEM

80 - 100 A	65 - 69 C+	55 – 59 D+
75 - 79 B+		50 – 54 D
70 - 74 B	60 - 64 C	0 - 49 E

COURSE POLICIES

ATTENDANCE

The instructor is of the view that, a student who does not attend lectures has not learnt anything and therefore not fit to take any form of exams. Therefore, students are advised to show up and participate actively in every lecture as agreed on the timetable or as re-arranged.

PARTICIPATION

Class participation is an important component of this course. Students are expected to actively participate by asking questions, contributing to discussions and make presentations as needed. There will be zero tolerance for passive passenger participants. In view of that, 10% of total grades is assigned to participation and attendance.

RAPPORT

Students are advised as a matter of urgency to prompt the instructor with regards to any challenges they experience with the course; this will enable the instructor to help find solution to the challenge as early as possible than to wait till the latter stages of the semester before making known the challenge.

PRESENTATION

The following presentation guidelines should be followed:

- The instructor will sit at the back of the class to evaluate the presenters. Make sure those at the back can read your slides and recognize your voice.
- You must practice your presentation orally as a group to ensure timing and transitions.
- You must make short bullet points that summarize an idea you will talk about.
- If you show it, talk about it. No point showing content that is not presented orally.
- Present an outline at the start of the presentation to set the expectations of the • audience.
- Project your voice with energy! Talk to the audience as opposed to your slides or paper.
- Although they are group presentations, marks will be allotted individually, therefore, proof of contribution to the work presented will be inspected.
- A presenter for the group will be chosen at random.

- All presentation slides must be submitted 10 mins before presentation begins unless otherwise indicated.

ACADEMIC DISHONESTY

The following practices constitute academic dishonesty and will result in an automatic F grade or other sanctions as deemed fit by the university's policy on academic dishonesty.

- Paper written by/for someone else
- Identical assignments by different students
- Purchasing, stealing, or borrowing the academic work of another